



PII: S0149-1970(02)00012-4

ENERGY DEPENDENCE OF RELATIVE ABUNDANCES AND PERIODS OF DELAYED NEUTRONS FROM NEUTRON-INDUCED FISSION OF ^{235}U , ^{238}U , ^{239}Pu IN 6- AND 8-GROUP MODEL REPRESENTATION

V.M. PIKSAIKIN, L. E. KAZAKOV, S. G. ISAEV, M. Z. TARASKO,

V. A. ROSHCHENKO, R. G. TERTYTCHNYI

Institute of Physics and Power Engineering, Bondarenko Sq.1, Obninsk, Kaluga region, Russia, 249020

G. D. SPRIGGS, J. M. CAMPBELL

Los Alamos National Laboratory, P.O. Box 1663, MS F664, Los Alamos, NM 87545-0001

ABSTRACT -The energy dependence of the relative abundances and periods has been measured for neutron induced fission of the main fuel nuclides, ^{235}U and ^{239}Pu in the energy range from epi-thermal to 5 MeV and ^{238}U in the energy range from 1 to 5 MeV. The efforts undertaken in improving the experimental and data processing procedures made it possible to improve the accuracy of the delayed neutron parameters determining their time-dependent behavior. In terms of the average half-life of delayed neutron precursors – the value that unequivocally characterizes the particular fissioning system – the real scale of the changes in the relative abundances and periods of delayed neutrons were determined. © 2002 Published by Elsevier Science Ltd.

INTRODUCTION

The kinetic behavior of a nuclear reactor below prompt critical is determined primarily by the group constants (i.e., the relative abundances, Y_i , and the decay constants, λ_i) of the delayed neutrons produced during the fission process. The group constants are not necessarily linked to any particular physical quantity, but rather, are constants in a convenient mathematical function that represents the aggregate behavior of more than 270 delayed neutron emitters following a fission event. The delayed neutron group constants, Y_i and λ_i , obtained in out-of-pile, aggregate measurements (Keepin et al., 1957) are normally used in most reactor physics calculations. However, there are two other approaches that can be used to derive the delayed neutron group constants for a particular reactor system. The first one is the *microscopic approach* in which the group constants are obtained from a detailed analysis of the neutron emission probability and fission product yield for each individual delayed neutron emitter in a particular reactor system. This microscopic approach has been used to derive the delayed neutron group constants for 43 fissioning systems ranging from ^{227}Th to ^{255}Fm (Brady and England, 1989). These results have been incorporated into the ENDF/B-V and VI libraries. The second approach is based on a variety of in-pile measurement techniques in which the effective group constants can be obtained by observing the transient behavior of a particular system.

As a rule the experimental data are presented in the traditional 6-group model approximation, which is widely used in reactor physics. But as it was shown by Spriggs et al. (1999) the traditional 6-group

model can be as well represented by an 8-group model based on a consistent set of precursors half-lives. In this study, 245 sets of experimentally measured 4-, 5-, and 6-group delayed neutron models were successfully expanded into an 8-group model which maintains the original reactivity scale and the time dependent behavior of the neutron population for positive reactivities. Besides this, an 8-group model, being more physically based, can correct known biases in the current reactivity scale in many existing reactors.

In spite of great efforts devoted to improving the time dependent delayed neutron data the problem of choosing the most adequate delayed neutron parameters set for a reactor with specific neutron spectrum still exists. It is a well known fact that the accuracy of any evaluated nuclear data depends on both the quality and quantity of experiments the nuclear data from which are considered in the evaluation process. Depending on the type of nuclear data, different criteria are used for the estimation of its reliability when the procedure of the data averaging is fulfilled. In case of the delayed neutron group constants until now there has been no correct procedure for averaging the data obtained in different experiments which could take into account the strong cross-correlation of the delayed neutron parameters originating from the least-squares fitting process. Moreover the relative abundances and decay constants are considered frequently as values independent from each other (Tuttle, 1975; Das, 1994) and on this basis the decay data set obtained for one fissioning system or averaged over a number of isotopes was used for constructing the delayed neutron parameter set for another isotope without appropriate adjustment of the fractional delayed neutron yields (Tuttle, 1975). As a result of this situation the recommended experimental set of delayed neutron parameters for major fissionable isotopes was based on only one experiment (Keepin et al., 1957). The reasons for this are the high quality of Keepin's data and the absence of a reliable procedure for averaging the appropriate data obtained in experiments with different experimental conditions such as background, different time sequence, data processing method etc.

Another problem in the development of a reliable delayed neutron data base is connected with our poor knowledge of the energy dependence of the delayed neutron parameters (Y_i , λ_i). In the main, information on time dependent delayed neutron data is related to thermal, fast (usually fission spectrum) and high energy (about 14 MeV) fission (Spriggs and Campbell, 1999). The absence of a criterion to distinguish between the thermal and fast delayed neutron parameter sets sometimes leads to the recommendation to use the fast delayed neutron data (as having the best accuracy) for any application (Tuttle, 1975). However, there are differences between the delayed neutron parameter sets even within the fast neutron spectrum range due to differences in the primary neutron spectra.

Till now testing of separate sets of delayed neutron group constants (Y_i , λ_i) has been carried out basically by means of a comparison of the dependence of a reactivity on the asymptotic period of a reactor, obtained on the basis of these constants, with the appropriate experimental data. Within the framework of such an approach Spriggs (1993), on the basis of experiments on a research reactor UARR, has shown that the group parameters for ^{235}U from Keepin et al. (1957), satisfactorily describe the kinetics of the reactor UARR and allow the prediction of the reactivity of the system with an accuracy of 3 %. On the other hand, reactor experiments have shown that ENDF/B-VI group parameters for ^{235}U underestimate the reactivity by 2 to 47% in the range from +0.80\$ to -0.80\$. Large discrepancies between the experimental and microscopic delayed neutron data sets are found for plutonium and americium isotopes (Isaev et al., 2000). Such results emphasize the need for careful testing of delayed neutron data obtained in the frame of the microscopic approach that is used as an alternative to the experimental approach for deriving the delayed neutron group parameters and the delayed neutron energy spectra.

We now present another approach for the validation of delayed neutron time dependent data sets and studying the energy dependence. It is based on using in the data analyses the value of the average half-life of delayed neutron precursors originating in the neutron induced fission of heavy nuclei

$$\langle T \rangle = \sum_{i=1}^K Y_i \cdot T_i,$$

where T_i is the half-life of i -th delayed neutron group ($T_i = \ln 2 / \lambda_i$). Being the combination of the delayed neutron parameters (Y_i, λ_i) obtained in the least-squares fitting procedure of aggregate decay curves this value has a simple physical sense and is not affected by the correlations which inevitably exist between the delayed neutron parameters. Also, the average half-life of delayed neutron precursors does not depend on the group model that is used in the least squares fitting procedure. In the work by Piksaikin and Isaev (1998) it was found that each fissionable nuclide at definite excitation energy has its own value of the average half-life of delayed neutron precursors. The dependence of the average half-life of delayed neutron precursors on the fissility parameter Z^2/A for fast neutron induced fission of isotopes of thorium, uranium, plutonium and americium is presented in Fig.1. Thus the average half-life value in addition to the dependence of reactivity on asymptotic period of a reactor can serve as a reliable criterion for the validation of the delayed neutron group parameters. Therefore, the information on the behavior of the systematics as a function of the excitation energy of fissionable nuclides is of great importance for reactor physics studies.

The first objective of the present work has been to conduct detailed measurements of relative abundances and half-lives of separate delayed neutron groups for neutron induced fission of the main fuel isotopes, ^{235}U , ^{238}U and ^{239}Pu , in the energy range from epithermal to 5 MeV. The second objective has been the processing of the experimental data in an 8-group model representation based on a consistent set of precursor half-lives (Spriggs et al., 1999).

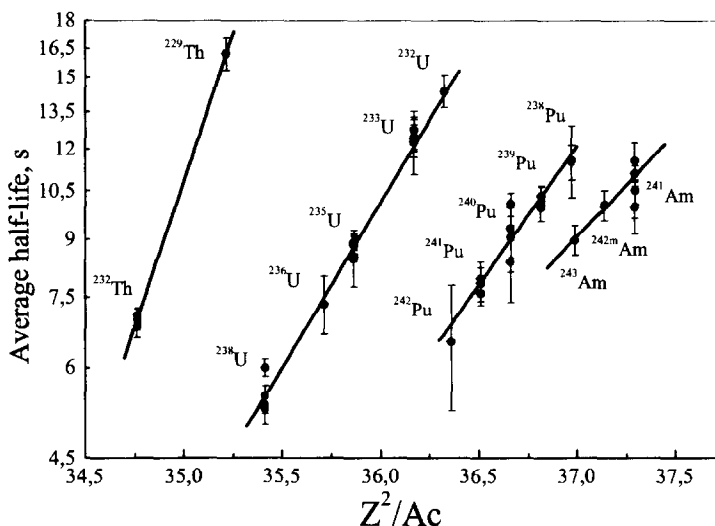


Fig.1 Systematics of the average half-life of DN precursors (group parameter values are taken from compilation by Spriggs and Campbell (1999)).

EXPERIMENTAL METHOD

The measurements were carried out on the beams of protons and deuterons at the cascade generator CG-2.5 of IPPE. Both fast monoenergetic and low energy neutrons were generated using the $\text{T}(p,n)^3\text{He}$ and $\text{D}(d,n)^3\text{He}$ reactions in solid tritium and deuterium targets. The epithermal neutron spectrum was obtained within a 20 cm cubic polyethylene block mounted near the accelerator target. A schematic diagram of the low energy experimental arrangement is shown in Fig.2. An appropriate neutron energy spectrum was calculated using the Monte Carlo program (Piksaikin, Shorin and Tertychnyi, 1999) and is presented in Fig.3. The average energy of slowing down neutrons averaged over the sample was 2.85 eV. In the fast neutron experiments the polyethylene block was removed and the pneumatic tube with fissionable sample was installed near the accelerator target. The fast neutron spectra corresponding to the $\text{T}(p,n)^3\text{He}$ neutrons and averaged over the fissionable sample are shown in Fig.4. The angle and energy dependence of the

neutrons from the $T(p,n)^3\text{He}$ and $D(d,n)^3\text{He}$ neutron sources as well as neutron multiple-scattering effects in construction materials was taken into account in the calculations of the neutron energy distribution averaged over sample volume (Piksaikin, Shorin and Tertychnyi, 1999). Thus the irradiations of the samples were made in well-defined neutron environments.

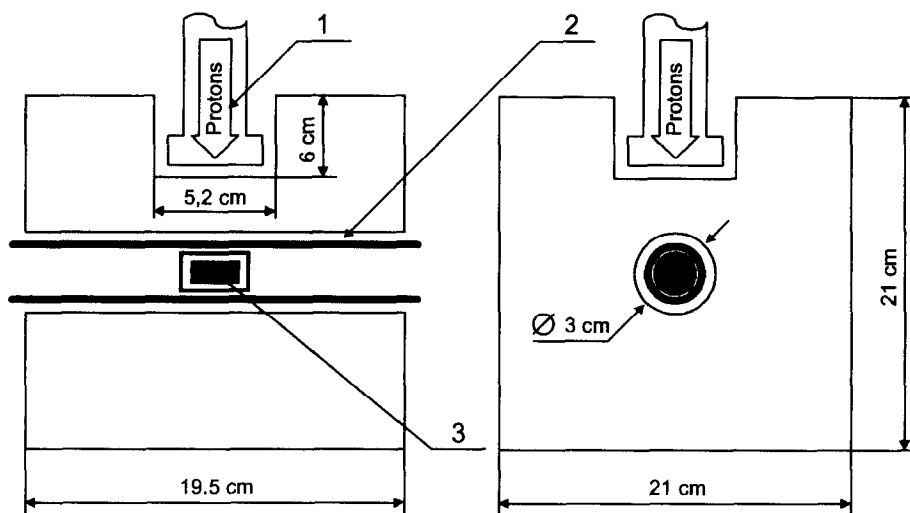


Fig. 2 Neutron source (accelerator target) and sample position for the low energy experiment.

1 – target assembly, 2 – tube of sample transfer system, 3 – fissionable sample.

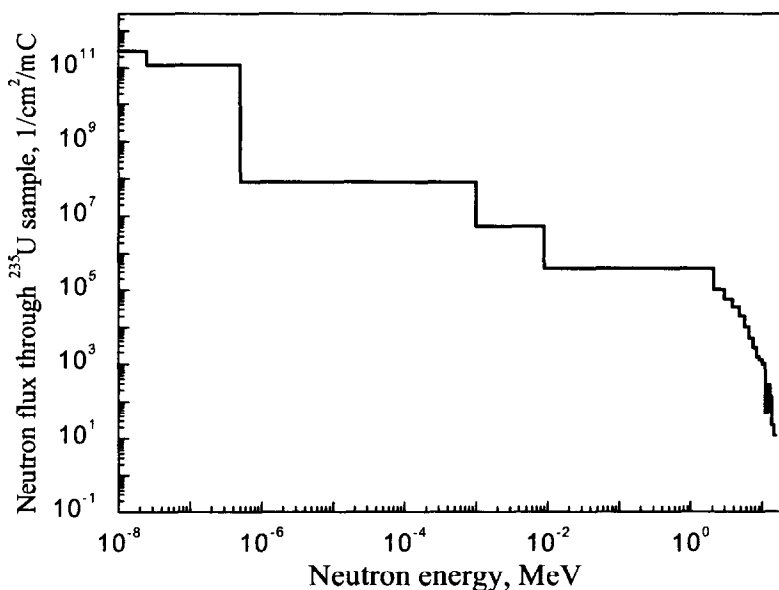


Fig. 3 Neutron energy distribution averaged over the sample under investigation.

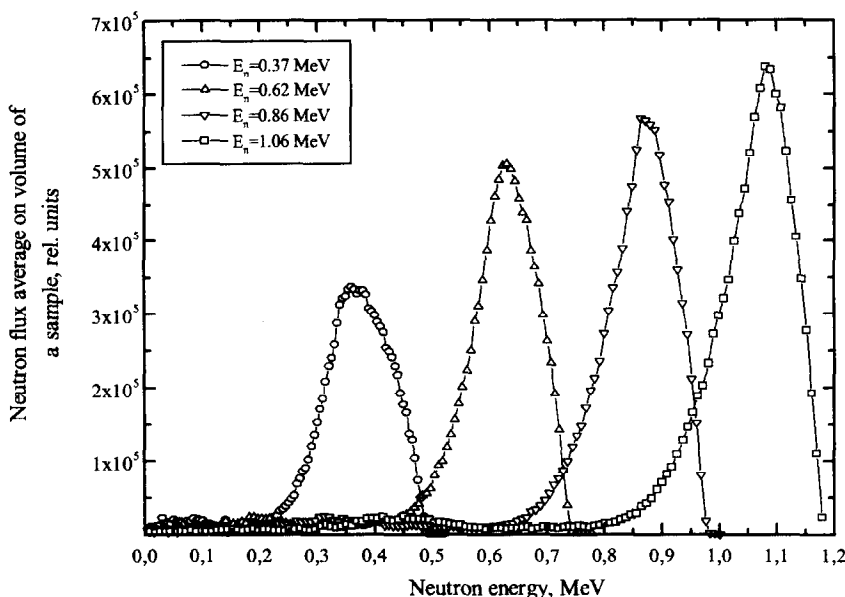


Fig.4 Fast neutron energy distribution averaged over the sample under investigation.

A boron counter of SNM-11 type was chosen as the main detector-counting unit working in the proportional mode operation. The detector is an assembly of 30 solid boron counters distributed in the polyethylene moderator along three concentric circles with diameters of 106, 160 and 220 mm. The detector is shielded against neutron background by borated polyethylene, boron carbide powder and cadmium sheets (Piksaikin, Korobeynikov et al., 1997). The pneumatic transfer system was used for transportation of a sample from the irradiation position to the neutron detector. The minimum sample delivery time was about 150 ms.

Data acquisition and processing system makes it possible to measure the following parameters: time dependencies of the neutron flux from the target and the ion current on the target of the accelerator, time dependence of delayed neutron intensity from irradiated samples, sample transportation time, and delay time of the sample at irradiation position. The PC computer serves as a central processor controlling the irradiation and counting time, the number and the width of the time channels for the delayed neutron intensity measurements.

The experimental method employed in the measurements was based on a cyclic irradiation of fissionable samples by neutrons with subsequent measurements of the time dependence of delayed neutron intensity (Piksaikin et al., 1998). The measurements with different irradiation and delayed neutron counting time intervals were designed to emphasize a certain delayed neutron group. The irradiation time intervals were 15, 180 and 300 s. The delayed neutron counting intervals were 424.5 and 724.5 s with the time channel sequence after the end of sample irradiation: 0.01 s (150 channels), 0.02 s (150 channels), 0.1 s (200 channels), 1 s (200 channels) and 10 s (30 or 50 channels). An example of the experimentally measured delayed neutron curve from fission of ^{235}U is presented in Fig.5.

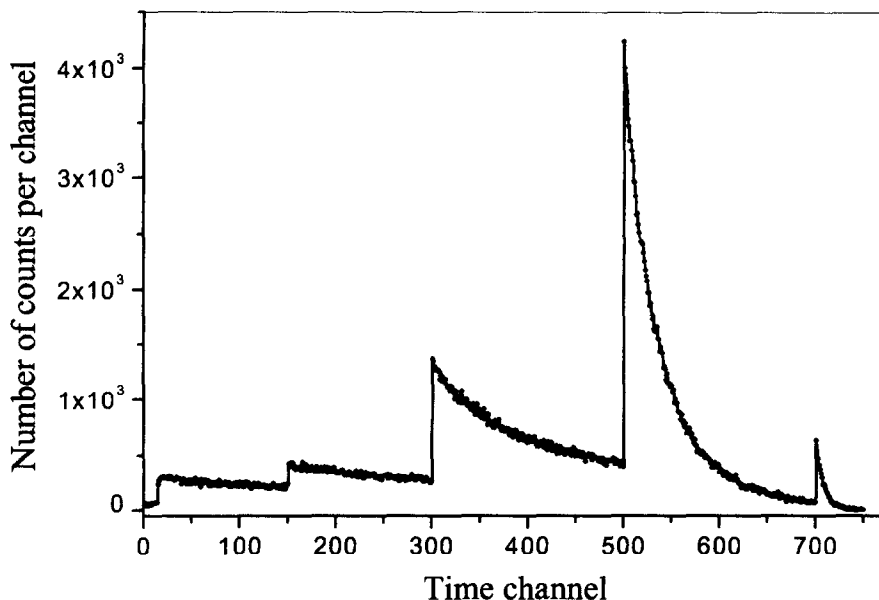


Fig. 5 Delayed neutron decay curve for the 1.01 MeV neutron induced fission of ^{235}U .

DELAYED NEUTRON DATA PROCESSING

The mathematical expectation of the number of counts z_i detected in the time interval (t_{i-1}, t_i) in the cyclic measurements with the irradiation time t_{irr} , N cycles each of which has duration T and the neutron background intensity β can be represented by the following expression

$$M(z_i) = \alpha \cdot \sum_{k=1}^K p_k \cdot \tau_k \cdot \left(1 - e^{-\frac{t_{irr}}{\tau_k}} \right) \cdot \left(1 - e^{-\frac{t_i - t_{i-1}}{\tau_k}} \right) \cdot e^{-\frac{t_{i-1} - t_{irr}}{\tau_k}} \cdot \left[\frac{N}{1 - e^{-\frac{T}{\tau_k}}} - \frac{e^{-\frac{T}{\tau_k}}}{\left(1 - e^{-\frac{T}{\tau_k}} \right)^2} \cdot \left(1 - e^{-\frac{N \cdot T}{\tau_k}} \right) \right] + \beta \cdot (t_i - t_{i-1}) \cdot N$$

where α - delayed neutron saturation activity, K - the number of delayed neutron groups, p_k - probability that precursors belong to the k -th delayed neutron group, τ_k - mean life-time of precursors corresponding to k -th delayed neutron group ($\tau_k = 1/\lambda_k = T_k/\ln 2$, λ_k and T_k being the decay constant and half-life of the k -th delayed neutron group, respectively). The numbers z_i can be considered as independent random variables with the mathematical expectation that depends on $n = 2 \cdot K + 2$ unknown parameters

$$\{\alpha, \beta, p_k, \tau_k, k = 1, \dots, K\} \quad (1)$$

From the physical point of view all parameters in set (1) must be non-negative. The p_k parameters should

be normalized to unity. Therefore it is suitable to make the following parameterization

$$p_k = \frac{y_k}{\sum_{j=1}^K y_j}$$

If one determines the value of one of the group y_k as equal to unity (usually this is the longest lived group of delayed neutron precursors) then the remaining y_k will have the sense of being the relative yields of delayed neutrons in respect to this group. Then the set (1) is transformed to the set

$$\{\alpha, \beta, y_k, \tau_k, k = 1, \dots, K\} \quad (2)$$

all elements of which are non-negative and one of the y_k is equal to unity. It is convenient to introduce for the set (2) a new designation a_j with renumbering them from 1 to n and considering them as the coordinates of vector $\vec{a}$. Having the explicit expression

$$M(z_i) = F_i(a_1, \dots, a_n)$$

one may formulate the task of the estimation of parameters $\{a_1, \dots, a_n\}$ as the search for an optimal description of the measured values z_i by the functions $F_i(\vec{a})$. The classical approach requires making an estimation of the parameters on the basis of a minimum of the function

$$Q(a) = \sum_{i=1}^m w_i \cdot [z_i - F_i(\vec{a})]^2,$$

where w_i – weighting factors which usually are taken as a value that is the inverse to the dispersion of z_i

$$w_i = [D(z_i)]^{-1}$$

However such an approach allows a strict statistical substantiation only for a function $F_i(\vec{a})$ with linear dependence on the parameters and when the parameters themselves have no restrictions.

In the general case one must use the iteration process based on the method of successive approximations (Keepin, 1957). In each step of this process the task of the linear estimation of the corrections to the existing parameters is solved. Statistically the iteration process can be formulated in the following manner. The hypothesis H_0 is raised that the values of the parameters are

$$\{a_j = a_j^{(0)}, j = 1, \dots, n\}$$

An alternative hypothesis H is that the parameters are presented in the following form

$$\{a_j = a_j^{(0)} \cdot e^{\delta_j}, j = 1, \dots, n\}$$

By expanding $F_i(\vec{a})$ in a Taylor's series in the vicinity of $\vec{\delta} = \{\delta_j = 0, j = 1, \dots, n\}$ the linear model in respect to the corrections $\vec{\delta}$ is created. This model is used for the estimation of $\vec{\delta}$ on the basis of measured values z_i . If the obtained corrections considerably differ from zero then the hypothesis H_0 is discarded and a

corrected hypothesis is put forward. If the difference of the corrections from zero is considered as negligible then the hypothesis H_0 is accepted and the vector $\bar{a}^{(0)}$ is chosen as the resultant estimation. The statistical uncertainties of the estimation are determined by the uncertainty of the correction term

Let us present $F_i(\bar{a})$ as

$$F_i(\bar{a}) = F_i(\bar{a}^{(0)}) + \sum_{j=1}^n c_{ij} \cdot \delta_j,$$

where

$$c_{ij} = \left. \frac{\partial F_i(\bar{a})}{\partial a_j} \cdot \frac{\partial a_j}{\partial \delta_j} \right|_{\delta_j=0}$$

According to the Gauss-Newton method the correction $\bar{\delta}$ can be obtained as the classical solution of the least squares method

$$\bar{\delta} = (C^T \cdot W \cdot C)^{-1} \cdot C^T \cdot W \cdot (\bar{z} - \bar{F}(\bar{a}^{(0)})),$$

where $\bar{z}$ and $\bar{F}(\bar{a})$ – vectors with coordinates z_i and $F_i(\bar{a})$ respectively, W – diagonal matrix of weighting factors, C – matrix with elements $\{c_{ij}\}$. As can be seen such an approach requires the inversion of the n -order matrix. For relatively small values of n (≤ 5) one can hope for a successful completion of the iteration process for such kind of correction $\bar{\delta}$. In the case of large matrix orders, n , for successful completion of the estimation process additional constraints should be applied to the corrections $\bar{\delta}$ as was done by Marquardt (1963).

The present approach in solving the problem is that the estimation of the vector $\bar{\delta}$ is determined by means of the sequence of estimations (the method of successive approximations is used)

$$\{\bar{\delta}^{(i)}, i = 1, \dots, m\},$$

each of which uses only one measurement z_i and the previous estimation. Particularly for each i one should find $\bar{\delta}^{(i)}$ from the condition of the minimum of the following quadratic form

$$Q(\bar{\delta}^{(i)}) = (\bar{\delta}^{(i)} - \bar{\delta}^{(i-1)})^T \cdot V_{i-1}^{-1} \cdot (\bar{\delta}^{(i)} - \bar{\delta}^{(i-1)}) + w_i \cdot [z_i - F_i(\bar{a}^{(0)}) - c_i^T \cdot \bar{\delta}^{(i)}]^2,$$

where V_{i-1}^{-1} – matrix that is the inverse to the covariance matrix V_{i-1} of the estimation $\bar{\delta}^{(i-1)}$, c_i^T – i -th row of matrix C . From the Gauss-Markov theorem about the estimation of the generalized least squares method corresponding to the minimum of the $Q(\bar{\delta}^{(i)})$ value, one can obtain the following formulas

$$\bar{\delta}^{(i)} = \bar{\delta}^{(i-1)} + \frac{z_i - F_i(\bar{a}^{(0)}) - c_i^T \cdot \bar{\delta}^{(i-1)}}{D(z_i) + c_i^T \cdot V_{i-1} \cdot c_i} \cdot V_{i-1} \cdot c_i,$$

$$V_i = V_{i-1} - \frac{(V_{i-1} \cdot c_i) \cdot (V_{i-1} \cdot c_i)^T}{D(z_i) + c_i^T \cdot V_{i-1} \cdot c_i}$$

A remarkable feature of the estimation procedure based on these formulas is that there is no need to invert the matrix and even the covariance matrix V_i is obtained and used in an explicit form.

Having the vector $\vec{b}^{(i)}$, components of which are the delayed neutron parameters $\vec{b}^{(i)} \equiv (b_j^{(i)}, j=1, \dots, 2 \cdot K) \equiv (Y_1^{(i)}, \dots, Y_K^{(i)}, T_1^{(i)}, \dots, T_K^{(i)})$ in a K -group model approximation and the appropriate covariance matrix V_j for each experimental run j one can obtain the resultant set $\vec{b}$ of the relative abundances and periods averaged over M independent measurements according to the following expression

$$\vec{b} = V(V_1^{-1} \cdot \vec{b}^{(1)} + V_2^{-1} \cdot \vec{b}^{(2)} + \dots + V_M^{-1} \cdot \vec{b}^{(M)}), \quad (3)$$

where $V = (V_1^{-1} + V_2^{-1} + \dots + V_M^{-1})^{-1}$ - the covariance matrix of the averaged set $\vec{b}$ of the delayed neutron parameters. The uncertainties of the averaged values of the relative abundances and periods of delayed neutrons were taken as the square root of appropriate diagonal elements of the covariance matrix V . Thus, this procedure takes into account the strong cross-correlation between the delayed neutron parameters derived from the least squares fitting of the decay curves.

The information contained in the covariance matrix V was used also in the calculations of the uncertainties of the average half-life values $\langle T \rangle$ of delayed neutron precursors

$$\langle T \rangle = \sum_{i=1}^K T_i Y_i$$

A 6-group model was used in the primary processing of the experimental data. Decay curves obtained in the experiments with the irradiation times of 180 and 300 s were used for the estimation of the delayed neutron parameters for the four longest groups. The parameters of the remaining two groups were estimated as a rule on the basis of the decay curves corresponding to experiments with the irradiation time of 15 s.

DATA ANALYSIS

As was noted in the introduction the comparison of the delayed neutron parameters (Y_i, T_i) obtained in the present work with appropriate data from other works was performed in terms of the average half-life $\langle T \rangle$ of the delayed neutron precursors. The same approach was used in determining the energy dependence of the relative abundances and periods of the delayed neutrons. It should be noted that the $\langle T \rangle$ value is not dependent on whether one considers K individual precursors or K delayed neutron groups of precursors nor on the temporal boundaries of the delayed neutron groups nor on cross-correlations between group parameters.

1. THE LOW ENERGY DATA

A summary of the results on the delayed neutron parameters (Y_i, T_i) for the low energy fission of ^{235}U and ^{239}Pu is given in Table 1. The average half-lives $\langle T \rangle$ of delayed neutron precursors based on these data are also presented in Table 1 for each of considered nuclides. For comparison purposes the $\langle T \rangle$ values derived from recommended experimental data of Keepin et al. (1957) and the data obtained using the microscopic approach by Brady and England (1989) are also included in Table 1. It is seen from the table that the present data are in excellent agreement with Keepin's data and considerably disagree with the data obtained in frame of the microscopic approach. It is seen also that the accuracy of the present results is higher than was obtained in the work of Keepin et al. (1957) especially for ^{239}Pu . The main reason for this is likely to be related to the better justified procedure used in the present work for averaging the data obtained in different experimental runs (see expression (3)). As was discussed above the main feature of

the proposed method is the account for cross-correlations existing between the delayed neutron parameters obtained by the least-squares method.

TABLE 1 – Relative abundances and periods of DN for neutron induced fission of ^{235}U and ^{239}Pu (epi-thermal neutrons with mean energy of 2.85 eV)

Isotope	i	Group number						Average half-life, s
		1	2	3	4	5	6	
^{239}Pu	Y_i	0.035	0.302	0.204	0.332	0.084	0.043	10.59±0.17
		±0.001	±0.007	±0.005	±0.006	±0.003	±0.002	10.69±1.11 [1]
	T_i	53.19	22.61	5.59	2.17	0.621	0.256	9.66 [2]
		±0.46	±0.11	±0.08	±0.04	±0.022	±0.009	
^{235}U	Y_i	0.038	0.211	0.197	0.396	0.132	0.026	8.98±0.11
		±0.001	±0.004	±0.004	±0.005	±0.004	±0.001	9.02±0.34 [1]
	T_i	53.95	22.34	6.40	2.26	0.494	0.179	7.90 [2]
		±0.28	±0.13	±0.08	±0.03	±0.017	±0.006	

[1] - (Keepin et al., 1957)

[2] - (Brady and England, 1989)

2. ENERGY DEPENDENCE OF RELATIVE ABUNDANCES AND PERIODS OF DELAYED NEUTRONS

The experimental results on the energy dependence of the relative abundances and periods of delayed neutrons, presented in a 6-group model approximation, from neutron induced fission of ^{235}U , ^{238}U and ^{239}Pu , are given in Tables 2, 3 and 4 respectively. The energy dependence of the relative abundances and periods of delayed neutrons is analyzed in terms of the average half-life of delayed neutron precursors $\langle T \rangle$ that is plotted as a function of the energy of the primary neutrons for ^{235}U , ^{238}U , and ^{239}Pu in Figs. 6, 7 and 8 respectively. The appropriate experimental data from other studies are also presented in the figures. References for these data can be found in the compilation work of Spriggs and Campbell (1999), corresponding to the name of the first author indicated in the figures.

It should be noted that almost all the data shown in the figures were obtained using reactor neutron fluxes. Therefore where the information was available the neutron energy values used for plotting these data are the mean energies obtained by averaging over the neutron energy distribution with fission cross section taken as the weighing factor (Tuttle, 1975). The energy dependence of the average half-life of delayed neutron precursors obtained in the present work for each isotope was approximated by a linear function $\langle T \rangle = a + bE_n$. The results of the regression analyses are presented in Table 5. The analyses were made for two energy intervals. The first interval includes all neutron energy points investigated in the work. The second one does not include the thermal energy points in the case of ^{235}U and ^{239}Pu and the 1 MeV point of the ^{238}U data. The results of the analyses corresponding to the full energy range of primary neutrons are shown in Figs. 6, 7 and 8 by the solid lines.

It is seen from Figs. 6, 7 and 8 that in all investigated cases (^{239}Pu , ^{235}U , ^{238}U) the value of the average half-life of delayed neutron precursors decreases as the energy of primary neutrons increases from thermal (or 1 MeV for ^{238}U) to 5 MeV. The energy dependences, $\langle T \rangle(E_n)$, obtained in the present work for ^{235}U and ^{239}Pu are in good agreement with the corresponding dependence that can be derived from the thermal and fast neutron energy data (Keepin, 1957). On the other hand, the present data, $\langle T \rangle(E_n)$, show a disagreement with the corresponding data measured by Maksyutenko et al. (see Spriggs and Campbell (1999) for a complete listing of these measurements). The energy dependence, $\langle T \rangle(E_n)$, obtained on the basis of these data for ^{239}Pu , ^{235}U and ^{238}U clearly shows the increase of the $\langle T \rangle$ value starting far below the threshold of the $(n, n'f)$ reaction. Though according to the systematics presented in Fig.1 some increase in the $\langle T \rangle$ value is expected at the neutron energies that are higher than the threshold of the $(n, n'f)$ reaction due to the fission of the compound nucleus having one neutron less.

TABLE 2 – Relative abundances and periods of DN for fast neutron induced fission of ^{239}Pu

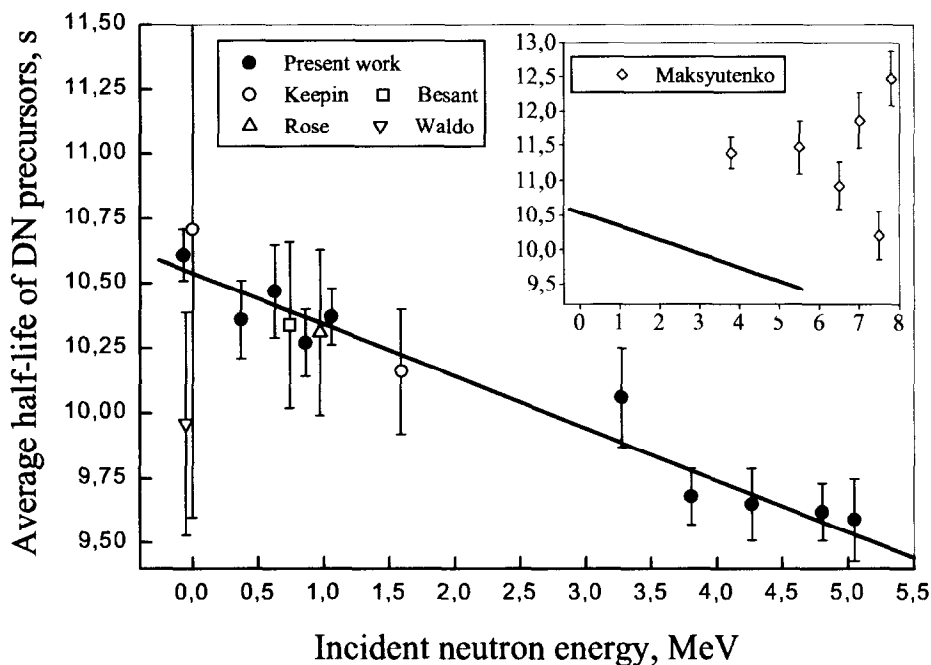
E_n , MeV		Group number						Average half-life, s
	i	1	2	3	4	5	6	
0.37 ± 0.06	Y_i	0.037 ± 0.001	0.295 ± 0.006	0.212 ± 0.005	0.319 ± 0.005	0.103 ± 0.003	0.034 ± 0.001	10.36 ± 0.15
	T_i	52.09 ± 0.41	22.34 ± 0.11	5.26 ± 0.07	2.06 ± 0.04	0.543 ± 0.015	0.218 ± 0.006	
0.62 ± 0.06	Y_i	0.037 ± 0.001	0.295 ± 0.007	0.217 ± 0.006	0.321 ± 0.007	0.096 ± 0.003	0.034 ± 0.001	10.47 ± 0.17
	T_i	53.19 ± 0.49	22.44 ± 0.12	5.30 ± 0.08	2.09 ± 0.04	0.568 ± 0.019	0.214 ± 0.008	
0.86 ± 0.06	Y_i	0.037 ± 0.001	0.288 ± 0.005	0.216 ± 0.005	0.321 ± 0.005	0.102 ± 0.003	0.034 ± 0.001	10.27 ± 0.13
	T_i	53.81 ± 0.41	22.19 ± 0.10	5.25 ± 0.07	2.06 ± 0.03	0.547 ± 0.016	0.217 ± 0.006	
1.06 ± 0.06	Y_i	0.038 ± 0.001	0.292 ± 0.004	0.214 ± 0.004	0.321 ± 0.004	0.101 ± 0.002	0.035 ± 0.001	10.37 ± 0.11
	T_i	53.51 ± 0.33	22.23 ± 0.09	5.11 ± 0.05	2.15 ± 0.03	0.555 ± 0.012	0.215 ± 0.005	
3.27 ± 0.14	Y_i	0.038 ± 0.001	0.271 ± 0.007	0.220 ± 0.006	0.331 ± 0.008	0.104 ± 0.003	0.035 ± 0.001	10.06 ± 0.19
	T_i	55.29 ± 1.45	22.23 ± 0.29	5.11 ± 0.13	2.16 ± 0.06	0.547 ± 0.016	0.216 ± 0.006	
3.81 ± 0.11	Y_i	0.038 ± 0.001	0.247 ± 0.004	0.227 ± 0.004	0.348 ± 0.005	0.105 ± 0.002	0.035 ± 0.001	9.68 ± 0.11
	T_i	56.19 ± 0.53	22.51 ± 0.14	5.22 ± 0.07	2.09 ± 0.03	0.552 ± 0.012	0.217 ± 0.005	
4.27 ± 0.11	Y_i	0.038 ± 0.001	0.243 ± 0.005	0.230 ± 0.005	0.347 ± 0.006	0.106 ± 0.003	0.035 ± 0.001	9.65 ± 0.14
	T_i	56.68 ± 0.65	22.49 ± 0.18	5.31 ± 0.08	2.10 ± 0.04	0.554 ± 0.016	0.217 ± 0.006	
4.81 ± 0.13	Y_i	0.039 ± 0.001	0.238 ± 0.004	0.228 ± 0.004	0.354 ± 0.005	0.105 ± 0.003	0.035 ± 0.001	9.62 ± 0.11
	T_i	57.49 ± 0.49	22.54 ± 0.14	5.32 ± 0.07	2.10 ± 0.03	0.554 ± 0.014	0.217 ± 0.005	
4.97 ± 0.13	Y_i	0.039 ± 0.001	0.243 ± 0.006	0.232 ± 0.006	0.341 ± 0.008	0.109 ± 0.004	0.036 ± 0.001	9.59 ± 0.16
	T_i	58.17 ± 0.65	22.04 ± 0.19	4.92 ± 0.09	2.17 ± 0.05	0.553 ± 0.019	0.217 ± 0.008	

TABLE 3 – Relative abundances and periods of DN for fast neutron induced fission of ^{235}U

E_n MeV	i	Group number						Average half-life, s
		1	2	3	4	5	6	
0.34 ± 0.07	Y_i	0.0370 ± 0.0005	0.227 ± 0.003	0.185 ± 0.003	0.400 ± 0.004	0.126 ± 0.003	0.0250 ± 0.0005	9.02 ± 0.07
	T_i	54.79 ± 0.19	21.80 ± 0.09	5.89 ± 0.06	2.293 ± 0.023	0.490 ± 0.010	0.180 ± 0.004	
0.59 ± 0.08	Y_i	0.0372 ± 0.0005	0.225 ± 0.003	0.189 ± 0.003	0.393 ± 0.004	0.130 ± 0.003	0.0258 ± 0.0005	9.03 ± 0.08
	T_i	54.57 ± 0.22	21.77 ± 0.11	5.96 ± 0.07	2.293 ± 0.028	0.494 ± 0.010	0.180 ± 0.004	
0.81 ± 0.09	Y_i	0.0360 ± 0.0004	0.224 ± 0.002	0.188 ± 0.003	0.398 ± 0.003	0.129 ± 0.002	0.0250 ± 0.0004	8.96 ± 0.06
	T_i	54.90 ± 0.16	21.86 ± 0.08	5.88 ± 0.05	2.232 ± 0.020	0.488 ± 0.009	0.180 ± 0.003	
1.01 ± 0.10	Y_i	0.037 ± 0.001	0.220 ± 0.004	0.192 ± 0.004	0.397 ± 0.005	0.129 ± 0.004	0.025 ± 0.001	8.91 ± 0.10
	T_i	54.72 ± 0.28	21.87 ± 0.13	5.86 ± 0.08	2.271 ± 0.029	0.499 ± 0.014	0.179 ± 0.005	
3.23 ± 0.15 one measurement	Y_i	0.038 ± 0.001	0.211 ± 0.007	0.194 ± 0.008	0.410 ± 0.011	0.122 ± 0.006	0.025 ± 0.001	8.87 ± 0.45
	T_i	56.51 ± 0.59	21.83 ± 0.29	5.95 ± 0.16	2.258 ± 0.065	0.520 ± 0.026	0.178 ± 0.009	
3.75 ± 0.14	Y_i	0.0370 ± 0.0005	0.204 ± 0.002	0.196 ± 0.003	0.412 ± 0.003	0.126 ± 0.002	0.0250 ± 0.0005	8.71 ± 0.06
	T_i	55.59 ± 0.18	22.00 ± 0.09	5.85 ± 0.05	2.261 ± 0.019	0.518 ± 0.010	0.179 ± 0.004	
4.20 ± 0.17 one measurement	Y_i	0.039 ± 0.001	0.205 ± 0.006	0.190 ± 0.008	0.407 ± 0.009	0.132 ± 0.006	0.027 ± 0.001	8.72 ± 0.37
	T_i	55.78 ± 0.43	21.70 ± 0.25	5.77 ± 0.13	2.252 ± 0.053	0.516 ± 0.025	0.179 ± 0.009	
4.72 ± 0.21	Y_i	0.0389 ± 0.0005	0.200 ± 0.003	0.187 ± 0.003	0.422 ± 0.004	0.126 ± 0.003	0.0261 ± 0.0005	8.63 ± 0.07
	T_i	55.49 ± 0.24	21.69 ± 0.11	5.84 ± 0.06	2.289 ± 0.021	0.499 ± 0.010	0.178 ± 0.004	

TABLE 4 – Relative abundances and periods of DN for fast neutron induced fission of ^{238}U

E_n MeV	i	Group number						Average half-life, s
		1	2	3	4	5	6	
1.01 ± 0.10	Y_i	0.0130 ± 0.0004	0.150 ± 0.004	0.168 ± 0.005	0.363 ± 0.008	0.229 ± 0.008	0.077 ± 0.002	5.81 ± 0.11
	T_i	52.43 ± 0.75	22.46 ± 0.21	5.18 ± 0.12	2.07 ± 0.05	0.467 ± 0.016	0.172 ± 0.006	
3.23 ± 0.15	Y_i	0.0130 ± 0.0002	0.135 ± 0.002	0.163 ± 0.003	0.389 ± 0.005	0.225 ± 0.004	0.075 ± 0.001	5.32 ± 0.05
	T_i	51.73 ± 0.35	21.75 ± 0.12	5.02 ± 0.06	1.972 ± 0.025	0.490 ± 0.010	0.172 ± 0.004	
3.75 ± 0.14	Y_i	0.0130 ± 0.0001	0.133 ± 0.001	0.164 ± 0.002	0.389 ± 0.003	0.227 ± 0.003	0.074 ± 0.001	5.31 ± 0.03
	T_i	52.91 ± 0.17	21.83 ± 0.06	5.12 ± 0.03	1.973 ± 0.013	0.496 ± 0.007	0.173 ± 0.003	
4.20 ± 0.17	Y_i	0.0130 ± 0.0002	0.130 ± 0.002	0.164 ± 0.003	0.385 ± 0.005	0.234 ± 0.005	0.074 ± 0.002	5.26 ± 0.05
	T_i	53.22 ± 0.26	22.01 ± 0.09	5.25 ± 0.05	1.925 ± 0.022	0.496 ± 0.011	0.175 ± 0.004	
4.72 ± 0.21	Y_i	0.0130 ± 0.0001	0.129 ± 0.001	0.166 ± 0.002	0.380 ± 0.003	0.237 ± 0.003	0.075 ± 0.001	5.28 ± 0.03
	T_i	53.39 ± 0.19	21.91 ± 0.06	5.12 ± 0.03	1.937 ± 0.014	0.481 ± 0.007	0.175 ± 0.003	

Fig.6 Energy dependence of average half-life of DN precursors for ^{239}Pu .

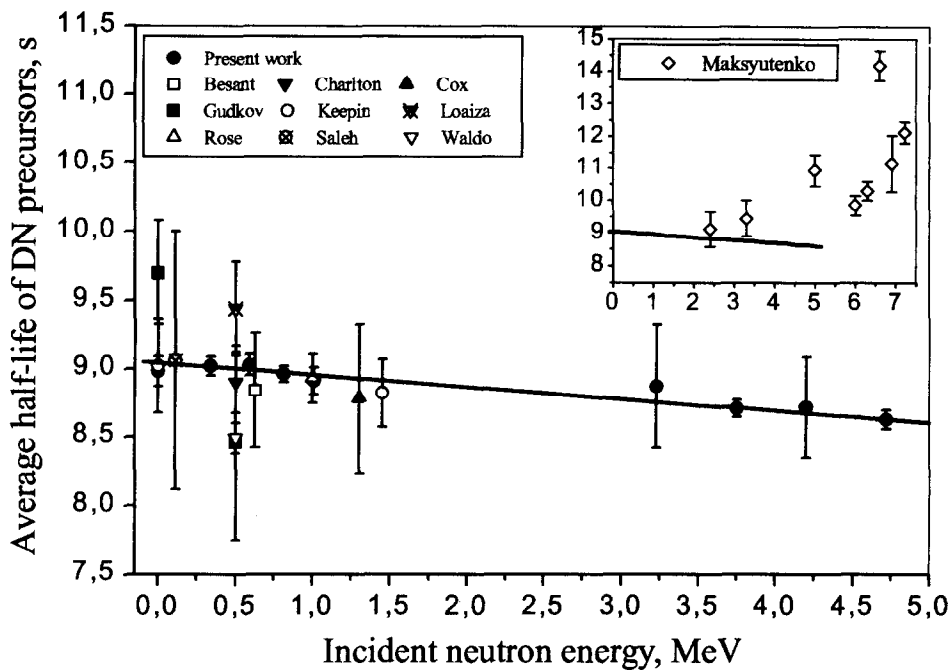
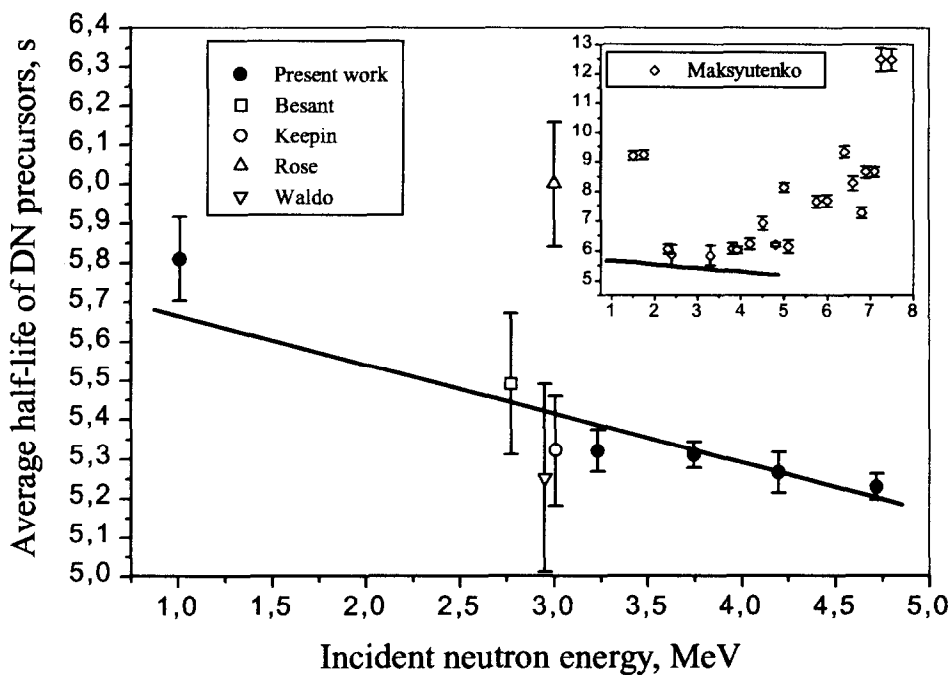
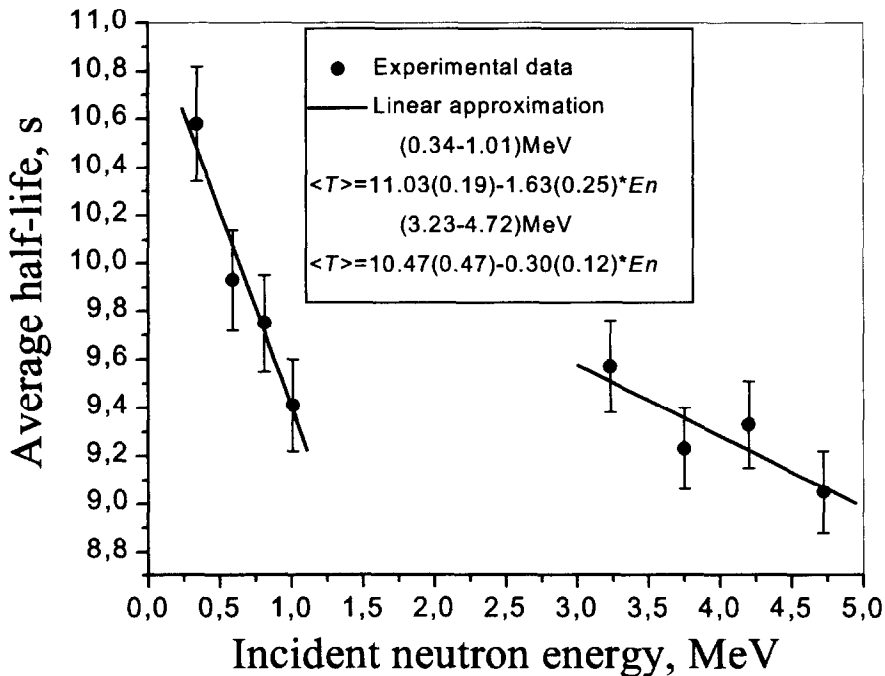
Fig.7 Energy dependence of average half-life of DN precursors for ^{235}U .Fig.8 Energy dependence of average half-life of DN precursors for ^{238}U .

TABLE 5 – Results of regression analyses of the delayed neutron group parameter data.

Isotope	Analyzed data	Intercept value A	Slope value B	Standard deviation
^{239}Pu	all data	10.55 ± 0.04	-0.201 ± 0.014	0.6883
	without the thermal value	10.50 ± 0.05	-0.190 ± 0.017	0.6615
^{235}U	all data	9.04 ± 0.01	-0.086 ± 0.005	0.3488
	without the thermal value	9.04 ± 0.01	-0.088 ± 0.005	0.3053
^{238}U	all data	5.79 ± 0.12	-0.125 ± 0.031	1.1972
	without the point below threshold	5.56 ± 0.05	-0.070 ± 0.012	0.3086

It is worth noting that although having the same general features, the $\langle T \rangle$ dependences of the considered nuclides differ from each other quantitatively. It is seen from Table 5 that in the case of ^{235}U and ^{239}Pu the intercept A and slope B practically does not change its value after removing the thermal point from the analyses. On the contrary in the case of ^{238}U the intercept and slope values, as well as the standard deviation value SD , are changed significantly if the 1 MeV point is excluded in the regression analysis. These results can be considered as the direct indication that the $\langle T \rangle(E_n)$ dependence for ^{238}U has irregularities at least in the sub-barrier energy region. An example of this kind of behavior of delayed neutron parameters can be observed in the case of the neutron induced fission of ^{237}Np (Piksaikin et al., 1998) (see Fig. 9). There are two energy regions with significantly different $\langle T \rangle(E_n)$ dependences. In the energy range of the (n,f) reaction threshold (0.34-1.01 MeV) the slope of $\langle T \rangle(E_n)$ data is 1.63 s/MeV. In the second region (3.23-4.72 MeV) the slope is 0.3 s/MeV. Taking into account the correlation existing between the total delayed yield and the average half-life of delayed neutron precursors (Piksaikin, Isaev, 1998) one can expect some irregularities (in the form of a departure from a constant value) in the energy dependence of the total delayed yield at the appropriate energy of incident neutrons as has already been observed in the case of ^{237}Np (Piksaikin, Shorin and Tertytchnyi, 1999).

Fig.9 Energy dependence of average half-life of DN precursors for ^{237}Np .

It is also worthwhile to note the importance of the different values of the slope of the $\langle T \rangle(E_n)$ dependence for ^{235}U and ^{239}Pu for reactor applications. As a matter of fact there is an empirical relationship between a relative change of the average half-life $\langle T \rangle$ of delayed neutron precursors and the appropriate relative change of reactivity ρ of the reactor system $\Delta\rho/\rho \approx \Delta\langle T \rangle/\langle T \rangle$, where the ρ value is inferred from the inhour equation using the same delayed neutron parameters as those used for the calculation of the $\langle T \rangle$ value. On the basis of a more pronounced decrease in the $\langle T \rangle(E_n)$ dependence for ^{239}Pu as compared with ^{235}U one can expect that the reactivity of a ^{239}Pu reactor system is much more sensitive to variations of the neutron energy spectrum than a reactor system composed of ^{235}U .

8-GROUP MODEL DATA

In the introduction it was noted that Spriggs et al. (1999) have developed a reliable procedure for expanding the experimentally-measured delayed neutron group constants into an 8-group delayed neutron model based on a consistent set of group half-lives. During the expansion process the following characteristics are conserved: the reactivity scale for positive reactivities, the time-dependent behavior of the reactor system, the mean half-life of the original delayed neutron set, as well as the uncertainties of the reactivity scale.

In the present work there was the intention to derive the 8-group delayed neutron constants from the experimental data without any constraints during the least squares fitting procedure. Two different approaches were taken. In one, the primary experimental delayed neutron decay curves were analyzed using the iterative least-squares procedure and a consistent set of decay constants from Spriggs et al. (1999). For each incident neutron energy a few sets of delayed neutron data were obtained. The final values of the relative abundances were obtained using the above-mentioned averaging procedure.

In the second approach the delayed neutron decay curves were generated using the 6-group data sets from Tables 2, 3 and 4. Then these decay curves were analyzed using the iterative least-squares method to obtain the relative abundances in the 8-group model representation. The relative abundance values in an 8-group model representation for neutron induced fission of ^{239}Pu , ^{235}U and ^{238}U obtained in these two approaches are presented in the Tables 6, 7 and 8 respectively. Comparison of the results made only in terms of the average half-life values, shows that the 8-group relative abundances derived on the basis of generated decay curves are in good agreement with the original 6-group data. The 8-group relative abundances obtained directly from the experimental data in some cases differ from those derived from the 6-group data mainly due to a poor statistics of the experimental data.

Summing up these results one may conclude with good reason that the 8-group generation approach developed by Spriggs et al. (1999) at present is the most reliable as it conserves as much of the information available in the original delayed neutron model as possible.

CONCLUSION

The energy dependence of the relative abundances and periods of delayed neutrons has been measured for neutron induced fission of the main fuel nuclides, ^{235}U and ^{239}Pu , in the energy range from epithermal to 5 MeV, and ^{238}U in the energy range from 1 to 5 MeV. The efforts undertaken in improving the experimental and data processing procedures have made it possible to improve the accuracy of the delayed neutron parameters determining their time-dependent behavior. In terms of the average half-life of delayed neutron precursors – the value that unequivocally characterizes the definite fissioning system – the real scale of the changes in the relative abundances and periods of delayed neutrons have been determined. It has been shown that for the neutron energy range up to the threshold of the (n,n'f) reaction the average half-life of delayed neutron precursors decreases when the energy of primary neutrons increases. This new proposed method of comparison of different sets of time dependent parameters of delayed neutrons has been proved to be an efficient tool for the development of a reliable delayed neutron data base. The obtained data can serve as a basis for the development of a delayed neutron data set appropriate for a particular reactor system with a specific neutron energy spectrum.

TABLE 6 – Relative abundances and periods of DN for fast neutron induced fission of ^{239}Pu in a 8-group model representation
(C – DN data obtained by conversion of the original 6-group model data;
E – DN data obtained in the analysis of the DN decay curve.)

E_n , MeV		Half-life, s								$\langle T \rangle$, s
		55.6	24.3	16.3	5.21	2.37	1.04	0.424	0.195	
		Group number								
		1	2	3	4	5	6	7	8	
$2.85 \cdot 10^{-6}$	C	0.029	0.240	0.083	0.188	0.289	0.089	0.059	0.023	10.63
		± 0.001	± 0.008	± 0.004	± 0.007	± 0.010	± 0.004	± 0.003	± 0.001	± 0.22
	E	0.029	0.237	0.087	0.182	0.305	0.077	0.061	0.022	10.62
		± 0.001	± 0.005	± 0.002	± 0.003	± 0.004	± 0.002	± 0.002	± 0.001	± 0.14
0.37 ± 0.06	C	0.028	0.236	0.078	0.182	0.284	0.097	0.071	0.024	10.37
		± 0.001	± 0.008	± 0.003	± 0.007	± 0.010	± 0.005	± 0.004	± 0.001	± 0.21
	E	0.028	0.229	0.082	0.176	0.290	0.072	0.099	0.024	10.23
		± 0.0004	± 0.005	± 0.0002	± 0.001	± 0.005	± 0.006	± 0.016	± 0.0005	± 0.13
0.62 ± 0.06	C	0.030	0.231	0.083	0.184	0.294	0.088	0.068	0.022	10.46
		± 0.001	± 0.008	± 0.004	± 0.007	± 0.010	± 0.004	± 0.003	± 0.001	± 0.22
	E	0.031	0.241	0.084	0.196	0.289	0.109	0.031	0.019	10.83
		± 0.0004	± 0.002	± 0.003	± 0.005	± 0.020	± 0.027	± 0.011	± 0.0004	± 0.09
0.86 ± 0.06	C	0.032	0.212	0.097	0.173	0.300	0.088	0.076	0.022	10.29
		± 0.001	± 0.007	± 0.004	± 0.007	± 0.010	± 0.004	± 0.004	± 0.001	± 0.20
	E	0.031	0.219	0.088	0.180	0.292	0.096	0.072	0.022	10.29
		± 0.001	± 0.007	± 0.008	± 0.009	± 0.007	± 0.002	± 0.002	± 0.0001	± 0.23
1.06 ± 0.06	C	0.032	0.220	0.089	0.172	0.322	0.061	0.084	0.020	10.38
		± 0.001	± 0.008	± 0.004	± 0.007	± 0.010	± 0.003	± 0.004	± 0.001	± 0.22
	E	0.032	0.228	0.094	0.176	0.335	0.026	0.084	0.025	10.68
		± 0.0002	± 0.004	± 0.004	± 0.009	± 0.014	± 0.014	± 0.010	± 0.002	± 0.13
3.27 ± 0.14	C	0.036	0.190	0.099	0.167	0.346	0.051	0.093	0.018	10.06
		± 0.001	± 0.007	± 0.004	± 0.007	± 0.011	± 0.003	± 0.005	± 0.001	± 0.20
	E	0.039	0.165	0.122	0.133	0.387	0.048	0.076	0.030	9.90
		± 0.002	± 0.011	± 0.007	± 0.002	± 0.005	± 0.005	± 0.008	± 0.003	± 0.31
3.81 ± 0.11	C	0.038	0.173	0.090	0.175	0.339	0.082	0.078	0.025	9.66
		± 0.001	± 0.006	± 0.004	± 0.007	± 0.010	± 0.004	± 0.004	± 0.001	± 0.18
	E	0.036	0.192	0.067	0.206	0.302	0.107	0.067	0.023	9.73
		± 0.002	± 0.008	± 0.007	± 0.011	± 0.024	± 0.015	± 0.007	± 0.001	± 0.27
4.27 ± 0.11	C	0.039	0.164	0.097	0.179	0.336	0.080	0.082	0.023	9.62
		± 0.001	± 0.006	± 0.004	± 0.007	± 0.010	± 0.004	± 0.004	± 0.001	± 0.18
	E	0.041	0.180	0.088	0.192	0.346	0.081	0.046	0.026	10.05
		± 0.001	± 0.004	± 0.005	± 0.011	± 0.027	± 0.020	± 0.019	± 0.003	± 0.16
4.81 ± 0.13	C	0.042	0.154	0.102	0.172	0.348	0.078	0.083	0.021	9.61
		± 0.002	± 0.006	± 0.004	± 0.007	± 0.011	± 0.004	± 0.004	± 0.001	± 0.20
	E	0.041	0.174	0.087	0.193	0.358	0.055	0.063	0.029	9.90
		± 0.002	± 0.010	± 0.010	± 0.005	± 0.009	± 0.018	± 0.015	± 0.005	± 0.32
4.97 ± 0.13	C	0.043	0.138	0.122	0.146	0.399	0.030	0.110	0.012	9.55
		± 0.002	± 0.005	± 0.004	± 0.006	± 0.011	± 0.002	± 0.005	± 0.001	± 0.18
	E	0.042	0.165	0.070	0.199	0.333	0.076	0.088	0.027	9.47
		± 0.003	± 0.025	± 0.025	± 0.035	± 0.042	± 0.017	± 0.004	± 0.001	± 0.78

TABLE 7 – Relative abundances and periods of DN for fast neutron induced fission of ^{235}U in a 8-group model representation

(*C* – DN data obtained by conversion of the original 6-group model data;

E – DN data obtained in the analysis of the DN decay curve.)

$E_n, \text{ MeV}$		Half-life, s								$\langle T \rangle, s$
		55.6	24.3	16.3	5.21	2.37	1.04	0.424	0.195	
		Group number								
		1	2	3	4	5	6	7	8	
$2.85 \cdot 10^{-6}$	C	0.034	0.153	0.086	0.212	0.298	0.105	0.073	0.039	9.00
		± 0.001	± 0.006	± 0.004	± 0.007	± 0.009	± 0.005	± 0.004	± 0.002	± 0.18
	E	0.036	0.143	0.118	0.182	0.362	0.032	0.102	0.025	9.32
		± 0.002	± 0.011	± 0.01	± 0.015	± 0.018	± 0.018	± 0.025	± 0.001	± 0.35
0.34 ± 0.07	C	0.034	0.146	0.104	0.176	0.361	0.056	0.098	0.025	9.04
		± 0.001	± 0.006	± 0.004	± 0.007	± 0.011	± 0.003	± 0.005	± 0.001	± 0.18
	E	0.033	0.148	0.105	0.180	0.352	0.076	0.079	0.027	9.06
		± 0.0003	± 0.0002	± 0.002	± 0.002	± 0.009	± 0.014	± 0.007	± 0.003	± 0.05
0.59 ± 0.08	C	0.034	0.145	0.104	0.184	0.346	0.062	0.099	0.026	9.03
		± 0.001	± 0.006	± 0.004	± 0.007	± 0.011	± 0.003	± 0.005	± 0.001	± 0.18
	E	0.034	0.145	0.100	0.196	0.318	0.060	0.122	0.025	8.97
		± 0.001	± 0.0004	± 0.001	± 0.004	± 0.009	± 0.021	± 0.023	± 0.001	± 0.07
0.81 ± 0.09	C	0.033	0.144	0.104	0.173	0.351	0.073	0.095	0.027	8.91
		± 0.001	± 0.006	± 0.004	± 0.007	± 0.011	± 0.004	± 0.005	± 0.001	± 0.18
	E	0.034	0.150	0.098	0.186	0.343	0.070	0.092	0.027	9.06
		± 0.0003	± 0.005	± 0.007	± 0.011	± 0.018	± 0.008	± 0.011	± 0.001	± 0.18
1.01 ± 0.10	C	0.034	0.144	0.099	0.183	0.352	0.067	0.096	0.025	8.94
		± 0.001	± 0.005	± 0.004	± 0.007	± 0.011	± 0.003	± 0.005	± 0.001	± 0.16
	E	0.035	0.159	0.097	0.212	0.335	0.120	0.032	0.010	9.46
		± 0.002	± 0.003	± 0.015	± 0.005	± 0.032	± 0.019	± 0.014	± 0.007	± 0.29
3.23 ± 0.15	C	0.039	0.121	0.115	0.175	0.368	0.071	0.087	0.024	8.91
		± 0.002	± 0.005	± 0.004	± 0.007	± 0.011	± 0.004	± 0.004	± 0.001	± 0.18
	E	0.039	0.128	0.108	0.201	0.318	0.167	0.023	0.016	9.05
		± 0.001	± 0.003	± 0.002	± 0.004	± 0.006	± 0.005	± 0.001	± 0.0005	± 0.10
3.75 ± 0.14	C	0.036	0.129	0.096	0.184	0.367	0.074	0.089	0.025	8.67
		± 0.001	± 0.005	± 0.004	± 0.007	± 0.011	± 0.004	± 0.004	± 0.001	± 0.16
	E	0.037	0.129	0.093	0.198	0.339	0.086	0.091	0.027	8.70
		± 0.001	± 0.003	± 0.002	± 0.004	± 0.005	± 0.002	± 0.003	± 0.001	± 0.10
4.20 ± 0.17	C	0.038	0.121	0.105	0.171	0.372	0.069	0.099	0.025	8.68
		± 0.002	± 0.005	± 0.004	± 0.007	± 0.011	± 0.003	± 0.005	± 0.001	± 0.18
	E	0.038	0.121	0.103	0.172	0.358	0.081	0.099	0.028	8.63
		± 0.001	± 0.002	± 0.002	± 0.003	± 0.004	± 0.002	± 0.003	± 0.001	± 0.08
4.72 ± 0.21	C	0.038	0.120	0.101	0.175	0.385	0.059	0.097	0.025	8.63
		± 0.001	± 0.005	± 0.004	± 0.007	± 0.011	± 0.003	± 0.005	± 0.001	± 0.16
	E	0.039	0.123	0.096	0.205	0.312	0.154	0.049	0.022	8.74
		± 0.001	± 0.002	± 0.008	± 0.0003	± 0.012	± 0.002	± 0.013	± 0.002	± 0.15

TABLE 8 – Relative abundances and periods of DN for fast neutron induced fission of ^{238}U in a 8-group model representation
(C – DN data obtained by conversion of the original 6-group model data;
E – DN data obtained in the analysis of the DN decay curve)

E = β -BN data obtained in the analysis of the BN decay curve)										
$E_n, \text{ MeV}$		Half-life, s								$\langle T \rangle, s$
		55.6	24.3	16.3	5.21	2.37	1.04	0.424	0.195	
		Group number								
		1	2	3	4	5	6	7	8	
1.01 ± 0.10	C	0.010	0.116	0.044	0.136	0.327	0.106	0.175	0.086	5.80
		± 0.0004	± 0.005	± 0.002	± 0.006	± 0.009	± 0.005	± 0.008	± 0.004	± 0.13
	E	0.006	0.116	0.045	0.138	0.324	0.105	0.194	0.072	5.60
		± 0.002	± 0.002	± 0.003	± 0.0004	± 0.001	± 0.036	± 0.007	± 0.029	± 0.14
3.23 ± 0.15	C	0.009	0.095	0.051	0.118	0.336	0.153	0.153	0.085	5.31
		± 0.0004	± 0.004	± 0.002	± 0.005	± 0.010	± 0.007	± 0.007	± 0.004	± 0.11
	E	0.010	0.101	0.048	0.128	0.344	0.160	0.106	0.103	5.52
		± 0.0001	± 0.005	± 0.004	± 0.009	± 0.017	± 0.024	± 0.017	± 0.006	± 0.15
3.75 ± 0.14	C	0.010	0.091	0.053	0.122	0.331	0.158	0.154	0.081	5.31
		± 0.0004	± 0.004	± 0.002	± 0.005	± 0.009	± 0.007	± 0.007	± 0.004	± 0.11
	E	0.010	0.095	0.054	0.135	0.336	0.170	0.163	0.037	5.52
		± 0.0001	± 0.001	± 0.002	± 0.004	± 0.006	± 0.005	± 0.014	± 0.008	± 0.05
4.20 ± 0.17	C	0.010	0.091	0.051	0.127	0.303	0.184	0.148	0.086	5.27
		± 0.001	± 0.004	± 0.002	± 0.005	± 0.009	± 0.008	± 0.007	± 0.004	± 0.12
	E	0.010	0.093	0.050	0.133	0.312	0.180	0.164	0.058	5.35
		± 0.0002	± 0.001	± 0.003	± 0.003	± 0.003	± 0.007	± 0.01	± 0.017	± 0.06
4.72 ± 0.21	C	0.011	0.088	0.051	0.123	0.316	0.161	0.162	0.088	5.24
		± 0.0005	± 0.004	± 0.002	± 0.005	± 0.009	± 0.007	± 0.008	± 0.004	± 0.11
	E	0.011	0.093	0.051	0.132	0.324	0.165	0.150	0.074	5.43
		± 0.0002	± 0.001	± 0.001	± 0.006	± 0.014	± 0.028	± 0.024	± 0.020	± 0.06

ACKNOWLEDGMENTS

It is a pleasure for the authors to thank Dr. J. Rowlands and Dr. A. D'Angelo for many helpful discussions, advice and help in preparation of the manuscript.

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